Zooming Through the Scale of Life

Learn.Genetics · University of Utah

Years 8-10	~30 minutes	Science · Biology

NOTICE — AI-generated worksheet
This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://learn.genetics.utah.edu/content/cells/scale/
Curriculum links: AC9S8U01 · AC9S8U02 · AC9S10U01

Learning intentions

- I can compare the relative sizes of cells, viruses, molecules and atoms using orders of magnitude.
- I can describe structures found inside a cell and link their size to their function.
- I can explain why DNA needs to be tightly packaged to fit inside a cell.

Getting started

Open the Learn.Genetics scale tool. Drag the slider slowly from left (largest, a coffee bean) to right (smallest, a carbon atom) at least once so you get a feel for the whole range before you start recording answers.

Tasks

1. Drag the slider until you reach a red blood cell. Name one other structure you pass close by on the scale, and estimate roughly how many red blood cells would fit across the width of a human hair.

2. Continue zooming in until you reach a typical virus. Is a virus larger or smaller than a red blood cell? Roughly how many times smaller or larger does it look?

3. Zoom to a cell nucleus, a virus, a DNA strand, and a carbon atom in turn. For each, estimate roughly how many of that structure would fit across the width of a human hair.

Structure	Roughly how many fit across a human hair's width

4. A plant cell and an animal cell both appear somewhere on this scale. Explain one specialised structure you would expect to find in a plant cell but not in a typical human cell, and how its size relates to its job.

5. Keep zooming in past the cell nucleus until you reach DNA. Describe what happens to the DNA's shape as you continue zooming in towards a single gene.

6. Explain why it is necessary for DNA to be coiled and packaged so tightly, given how small a cell nucleus is compared to the length of DNA if it were stretched out.

7. Zoom all the way to a carbon atom. Name one other structure on the scale that is roughly the same order of magnitude (size) as a carbon atom.

8. Structure and function are linked at every scale in this tool, from whole cells down to atoms. Choose one structure you zoomed past and explain how its size suits the job it does.

Extension (optional)

Estimate roughly how many carbon atoms placed end-to-end would be needed to span the width of a single human cell, using the scale markers in the tool to help you reason it out.